

IMAGE INPAINTING

Group 15 - Project 3
Faizuddarain Syam (1601040)
Junaid Mundichipparkkal(1553771)

1. Introduction

Image inpainting is the process of filling up the missing parts of the image which are distorted or missing due to various reasons. It can also be used to remove unwanted objects from the image. The applications of image inpainting have high importance in the computer vision fields such as on autonomous vehicles and photography. In this paper, the goal is to be able to perform image inpainting using convolutional neural networks, where the network will be trained and tested with a dataset of choice. The goal is to achieve the models with the best performance-based metrics such as Mean-Squared Error (MSE) Loss, Peak Signal to Noise Ratio (PSNR), and Structural Index Similarity (SSIM) values.

2. Problem Description

To inpaint an image there are two necessary inputs that are required. First is the image that needs to be inpainted. This image is then used as a ground truth to an input data. The second is a mask matrix that is the same size as the image representing areas in the input image that needs to be masked out. The result should be a modified image where the masked area is replaced by an image that is as close as possible to the ground truth.

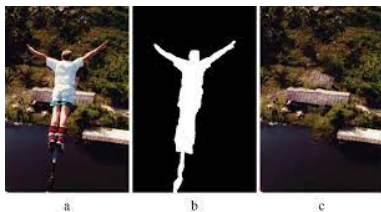


Figure 1. Example for image inpainting

For example Figure 1 (a) shows the input image, Figure 1 (b) shows the mask, and Figure 1 (c) shows the expected output from the neural network. Here, dataset used consists of normal images with random masks generated for each image such as (rectangle or free hand drawing) to train a convolutional neural network. The inpainting model training is first done for images of the face, then high resolution face images, and then street or landscape images.

3. Related work

There are mainly two types of image imprinting options namely patch matching and generative model. Patch matching is the process of finding a patch from another low-

resolution image to cover the mask area. Generative modeling is the approach that we focus on in this paper, where we use Convolutional neural network (CNN) models to synthesize the missing part of the image using the semantics learned from large datasets. The generative models have several approaches now to model different ways to inpaint images. The basic approach used in convolutional neural networks is to have one single input where the mask is added as a single new channel and the model is planning to learn from encoder and decoder based CNN architecture (Can use multilevel wavelets too). There are some new approaches such as partial convolutions and gated convolutional where the mask and distorted images are completely separate input and they follow a specific function to the masks, where the masks gradually will become less and less in the encoder part of CNN architecture. Both the methods are highly effective while gated uses binary-based masked dissolution while partial have a soft value based mask dissolution.

4. Proposed methodology

4.1. Method

The approach to building a CNN based model is started with the understanding of auto encoders. The UNet architecture is implemented with the input image that have a mask and the output image which is an original image. In this proposed methodology, various models we be evaluated based on two approaches towards finding the best model: the network architecture and the loss functions used. During the training process, a random rectangle will be generated as a mask for each input image and the corresponding input image image will be erased (set to 0) on the masked area.

4.2. Architecture

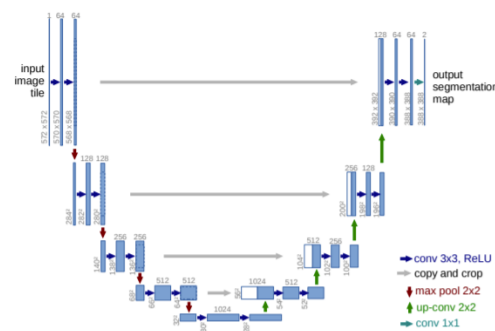


Figure 2. Unet architecture

Figure 2 shows the architecture of the UNet model which

is used. The main changes for the model can be classified into two, based on input type and based on the loss function.

Approach 1(a): The generated rectangle area within the image will erase the area of the image, setting the values for each RGB layers into 0. This means the the UNet model will only need a 3 layer input.

Approach 1(b): In addition to erasing the masked area of the input image, the generated rectangle matrix will be added as the fourth layer to input, resulting in a 4 layer input for the model.

Approach 2 (a): The loss of the model is simply a MSE (L2) loss. Which is propagated backwards to update the weights in the network.

Approach 2(b) : The loss of the model is a combination of both L2 loss and PSNR value. The L2 loss is performed separately on the area inside the mask and outside the mask. The PSNR value will be converted into a loss value where it will be reduced from its maximum value of 100 and normalized. The formula for each loss can is shown as follow:

$$\text{loss-in} = \text{MSELoss}((1 - M) * Y_{pred}, (1 - M) * Y_{true})$$

$$\text{loss-out} = \text{MSELoss}((M) * Y_{pred}, (M) * Y_{true})$$

$$\text{loss-PSNR} = (100 - \text{PSNR})/100$$

$$\text{Final-loss} = 600 * \text{loss-in} + 100 * \text{loss-out} + \text{loss-PSNR}$$

The Final-loss is then used to backpropagate in each epoch to update the model.

4.3. Data Description

The face image dataset has identical features for the convolutional layers extracted while the landscape or street view images have several features (since there will be objects of different types). So in order to develop and understand different model architectures, we will use the face dataset after we will scale our model to work on other datasets. First, for the face dataset, we will use low-resolution images of the celeb dataset, and then moving to the next part will be using the high-resolution face dataset from Flickr and for the final stage we will be building the model on the Cityscape dataset.

4.4. Evaluation metrics

4.4.1 Mean-squared Error (MSE) / L2 Loss

Loss based on squared value of difference between true and predicted values. Most commonly used. This loss helps to get a pixel to pixel comparison of the input and output of the model. It is also called MSE loss.

4.4.2 Peak signal to noise ratio (PSNR)

PSNR is most commonly used to measure the quality of reconstruction of lossy compression codecs. Since we are using auto-encoding based architecture in the training of the model it will be ideal to use PSNR as a metric.

4.4.3 Structural index similarity (SSIM)

SSIM quantifies image quality degradation caused by processing such as data compression or by losses in data transmission.

5. Current Status

Based on the approaches mentioned, we resulted in 4 models to train and evaluate, where:

Model 1 follows approach 1(a) for input and loss as approach 2(a). Model 2 follows approach 1(a) for input and loss as approach 2(b). Model 3 follows approach 1(b) for input and loss as approach 2(a). Model 4 follows approach 1(b) for input and loss as approach 2(b).

5.1. Results and Analysis

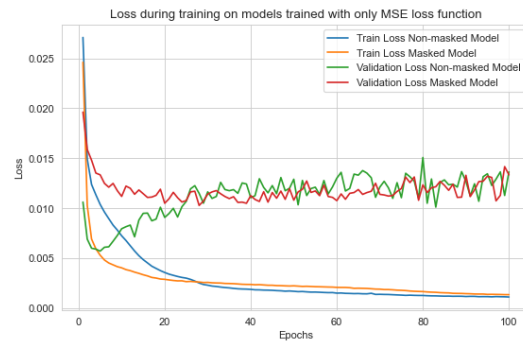


Figure 3. Training Loss for Model 1 and Model 3



Figure 4. Training Loss for Model 2 and Model 4

The Figure 5 shows results of different models which are implemented based on the approaches mentioned in section 4.2.

Comparing the image output results, Model 1 and Model 3, which both are based on approach 2(a), failed to be able to show details within the masked area. On the other hand, Model 2 and Model 4, which uses approach 2(b) as the loss function, were able to show more details. This shows that whichever approach 1 used (3 layers or 4 layers input), the

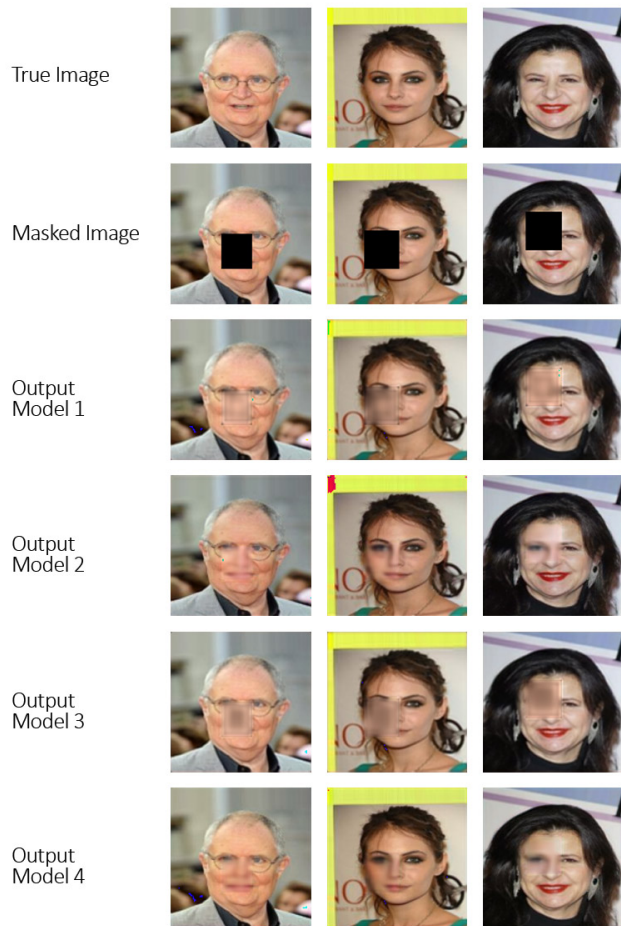


Figure 5. Image output results of different models

difference is shown by the loss function used during the training.

Figure 3 and Figure 4 shows the training loss and validation loss for all models. It can be seen that all the four models are learning. But when we use the MSE Loss the value of loss becomes small very fast leading to vanishing weights. Hence by comparing to Figure 5 it also supports our conclusion that approach 2(b) based loss is better for image inpainting.

Table 1 shows the three different loss functions mentioned in section 4.4. Here the image part represents the part of the image which is used for finding the loss. "All" represents the entire image, "Inside mask" represent the loss comparing only the values inside the masked area and vice versa for the "Outside mask".

As we can see, based on the output images before, Model 2 which suggested to show better results, falls short compared, or very similar, to Model 1 and Model 3 if we look at the image as a whole. However, comparing Model 1 and

Loss Function	Image Part	Model 1	Model 2	Model 3	Model 4
PSNR	All	77.6979	76.6544	76.8552	78.8502
	Inside Mask	78.4639	82.8909	77.5200	82.6876
	Outside Mask	86.7906	78.1559	86.3216	81.7158
SSIM	All	0.9602	0.9689	0.9607	0.9770
	Inside Mask	0.9781	0.9886	0.9758	0.9883
	Outside Mask	0.9892	0.9844	0.9921	0.9914
MSE Loss	All	0.00130	0.00149	0.00155	0.00092
	Inside Mask	0.00112	0.00042	0.00137	0.00044
	Outside Mask	0.00017	0.00107	0.00017	0.00047

Table 1. Evaluation results of different models

3 to Model 2 and 4, we can see on the image part inside the mask, the values show high differences, where Model 2 comes out to perform better.

On the other hand, looking at the outside mask losses, the values are still not good, showing that the model performs unnecessary adjustments to areas outside the mask. In conclusion to this problem, Model 4 performs better than Model 2.

The next step for this project will be

- Using a different and more complex dataset.
- Building masks of different shapes and hand drawings.
- Changing the architecture of the given model to suit the problem better with the help of hyperparameter tuning to get a better model architecture.
- Formulating a better total loss function.

6. Expected Challenges

- The result image should have a smooth transition while removing the mask. The model have to be trained with large set of images with more number of epochs might help us to get other this situation.
- The filled area should be more detailed and accurate to the true image. The idea is to try using a combination dropout and batch normalisation to help us to get through this challenge.
- The unmasked area of the model should not be affected while inpainting. This challenge can be solved with the help of using a partial convolution where the mask will be reduced gradually by later closing the whole mask area.
- The model and the process of training and testing should be able to perform better on larger sized image inputs.